Analytic Multi-Loop Helicity Amplitudes in QCD

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The full physical information of a color-ordered helicity amplitude is in the finite remainder:

$$\mathcal{R}^{\ell-\text{loop}}_{n-\text{point}}(|i\rangle_\alpha, [i|_{\dot\alpha}) = \sum_i r_i(|i\rangle_\alpha, [i|_{\dot\alpha})\, G_i(|i\rangle_\alpha \times [i|_{\dot\alpha}) \,.$$

Catani ('98)
Becher, Neubert ('09)
Gardi, Magnea ('09)
Weinzierl ('11)

Challenges:

- Compute the G_i , a basis of transcendental functions, e.g. the *Pentagon Functions*;

Chicherin, Sotnikov ('20) + Zoia ('21) + Abreu, Ita, Page, Tschernow ('23)

- Reconstruct the r_i , functions in the field of fractions over a polynomial quotient ring,

Peraro ('16); GDL, Maitre ('19); GDL, Page ('22).

$$R_n = \mathbb{F}\Big[|1\rangle_\alpha, [1|_{\dot\alpha}, \ldots, |n\rangle_\alpha, [n|_{\dot\alpha}\Big]\Big/ J \qquad \text{with} \qquad J = \text{Ideal}\Big(\textstyle\sum_{i=1}^n |i\rangle[i|\Big) \,,$$

in connection with algebraic geometry, and extended to massive particles (spin-spinors).

Arkani-Hamed, Huang, Huang ('17); Ochirov ('18); Shadmi, Weiss ('18); Campbell, GDL, Ellis ('25); GDL, Melnikov, Tresoldi ('26)

Amplitudes we computed: GDL + [Abreu, Campbell, Ellis, Ita, Klinkert, Kuschke, Maitre, Page, Ruf, Seth, Sotnikov, ...]

Inspire	Process	Loops	Year	Inspire	Process	Loops	Year	Inspire	Process	Loops	Year
1728738	$0 \to gggggg$	1L FC	2019	2060823	$0 \to q\bar{q}\ell\bar{\ell}\ell'\bar{\ell}'g$	1L FC	2022	2915977	$q\bar{q} \to t\bar{t}H$	1L FC	2025
1779473	$pp \to Hjj$	1L FC	2020	2663067	$pp \to \gamma\gamma\gamma$	2L FC	2023	2954479	$q\bar{q} \to HHH$	1L FC	2025
1826610	$pp \to jjj$	2L LC	2020	2723256	$pp \to jjj$	2L FC	2023	3128446	$pp \to Vjj$	2L LC	2026
1861752	$pp \to W(\ell\nu)\gamma j$	1L FC	2021	2821186	$pp \to HHj$	1L FC	2024	3152032	$pp \to Hjj$	2L LC	2026
2048108	$pp \to \gamma\gamma\gamma$	2L LC	2022	2899891	$pp \to V(\ell\bar{\ell})jj$	2L LC	2025				

Least common denominator degrees of freedom $pp \to \gamma\gamma\gamma$ 30k; jjj 250k ; Vjj 25M; Hjj 80M.